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NARUSE et al.(10) **Pub. No.: US 2025/0274019 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **ELECTRIC MOTOR AND MOTOR-DRIVEN  
COMPRESSOR****Publication Classification**(71) Applicant: **KABUSHIKI KAISHA TOYOTA  
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(2016.01); *H02K 11/27* (2016.01)(73) Assignee: **KABUSHIKI KAISHA TOYOTA  
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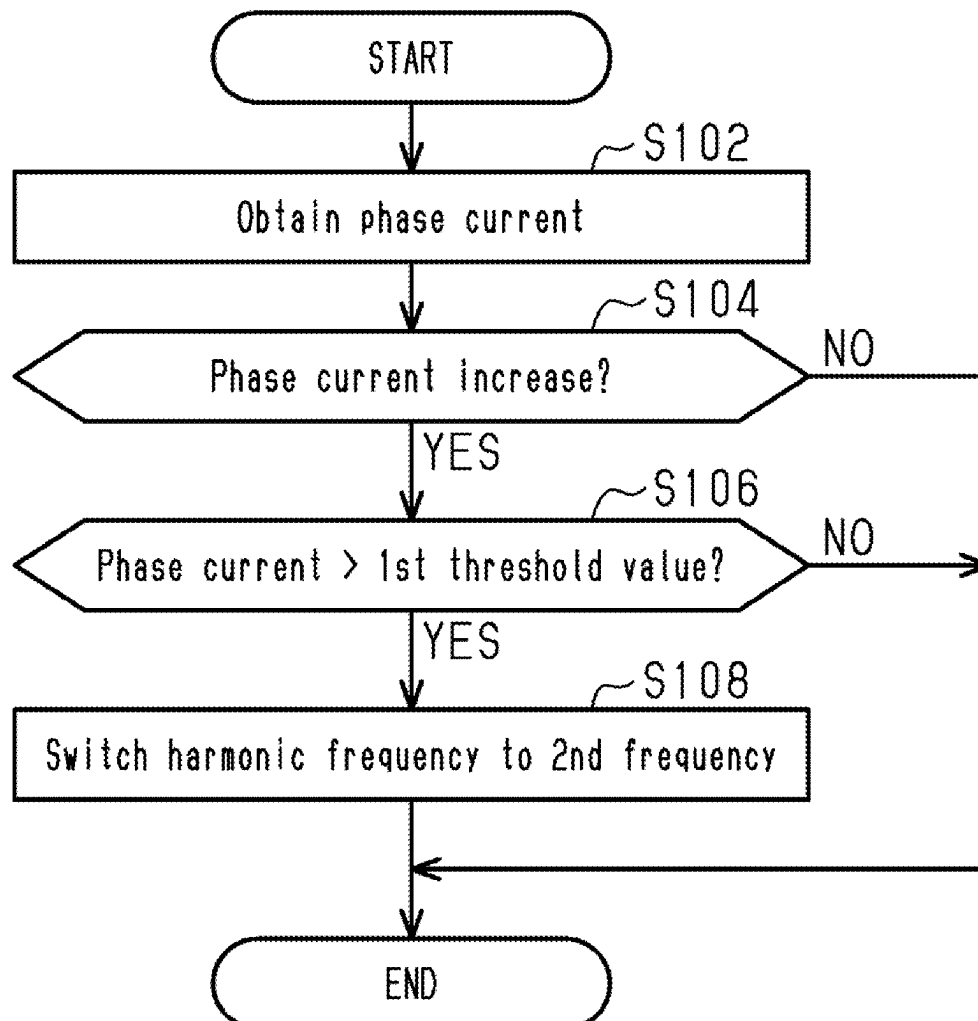
**ABSTRACT**

An electric motor includes processing circuitry. The processing circuitry is configured to superpose a harmonic on at least one of instruction values, and estimate the position of a rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values. The frequency of a harmonic includes a first frequency and a second frequency that is lower than the first frequency. The processing circuitry is further configured to determine whether a temperature-associated value becomes greater than a threshold value. When the temperature-associated value becomes greater than the threshold value, the processing circuitry is further configured to set the frequency of the harmonic for superposition on the at least one of instruction values to the second frequency.

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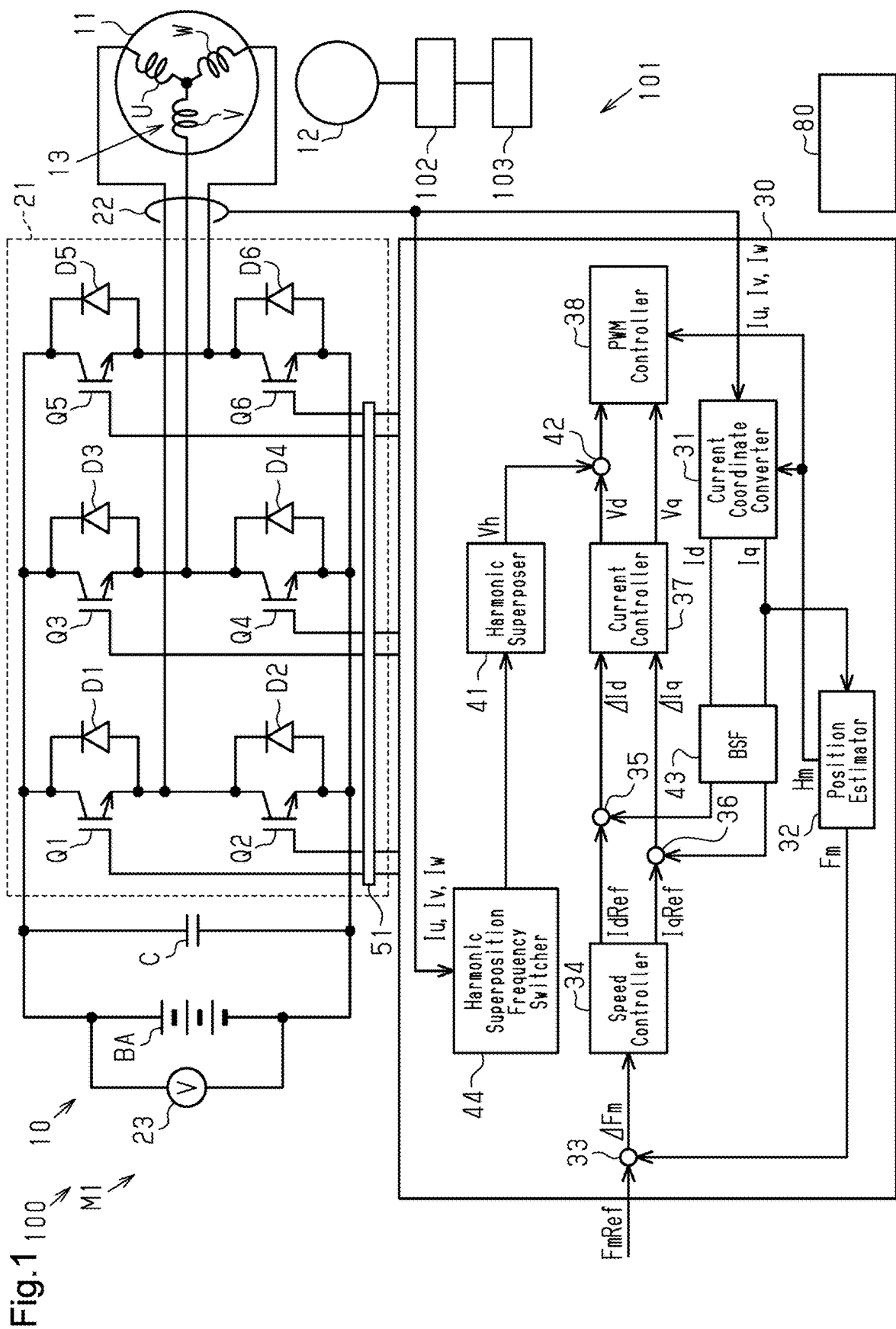


Fig.2

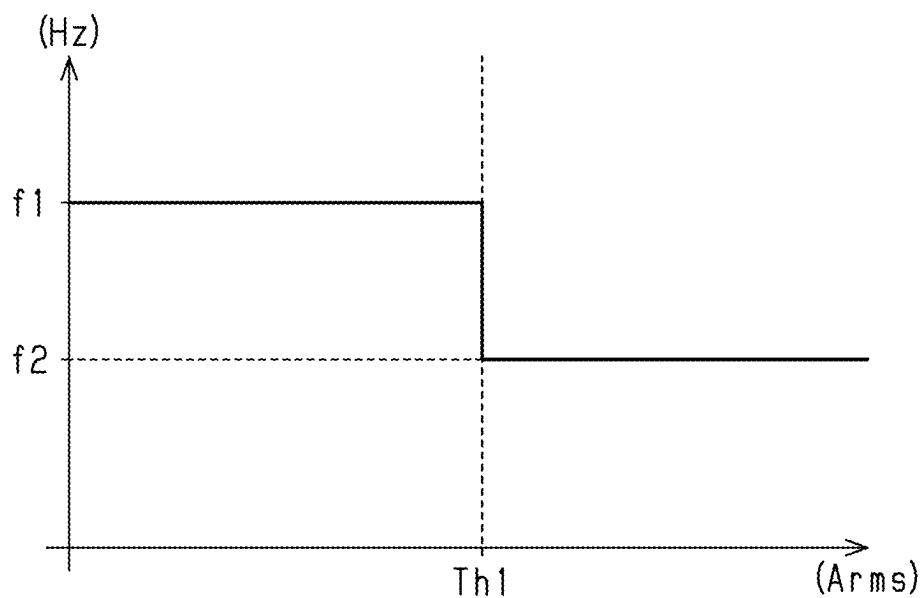


Fig.3

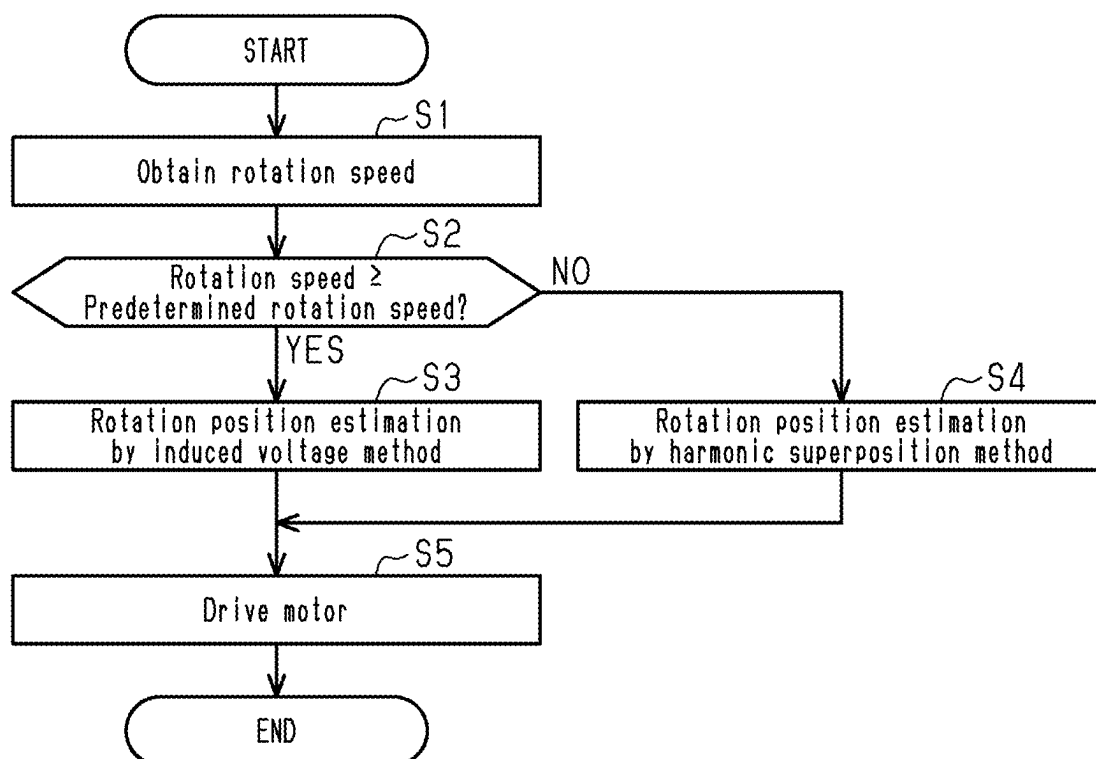


Fig.4

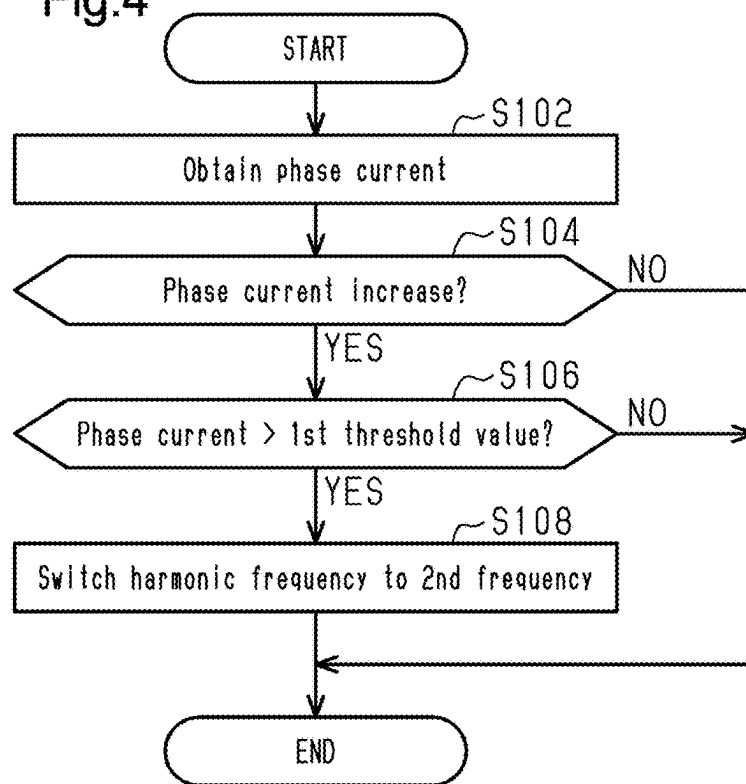


Fig.5

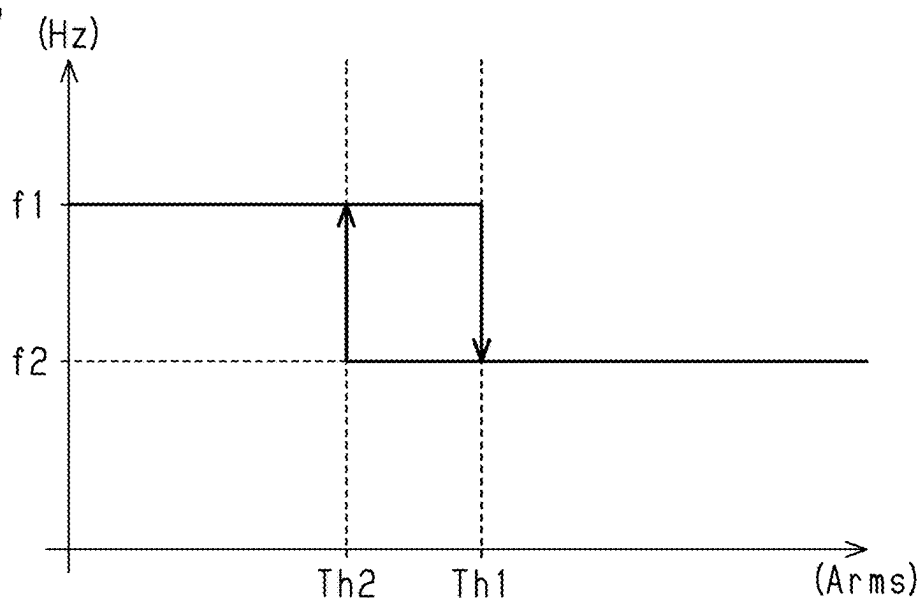
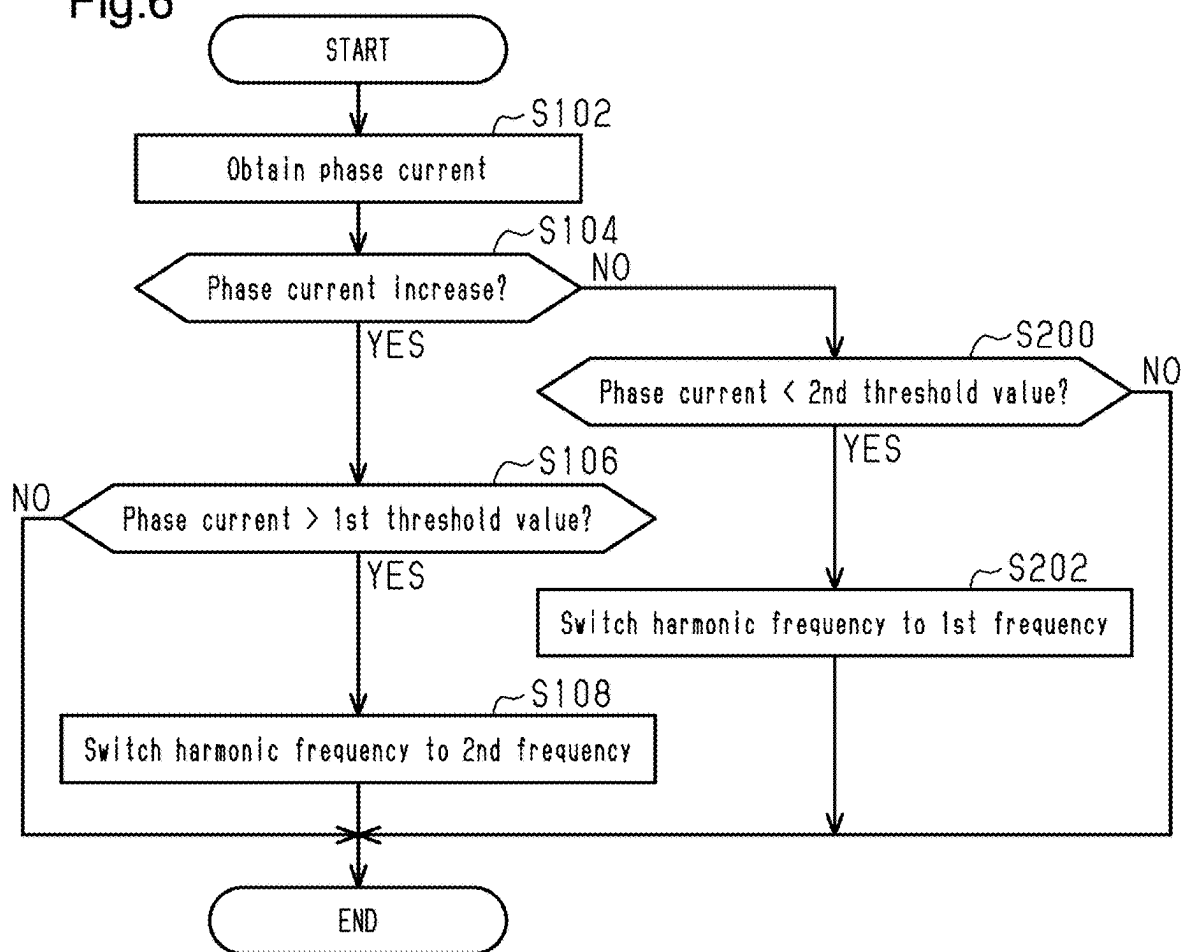


Fig.6



## ELECTRIC MOTOR AND MOTOR-DRIVEN COMPRESSOR

### CROSS-REFERENCE TO RELATED APPLICATIONS

**[0001]** This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-027574, filed on Feb. 27, 2024, and Japanese Patent Application No. 2024-228939, filed on Dec. 25, 2024, the entire contents of which are incorporated herein by reference.

### BACKGROUND

#### 1. Field

**[0002]** The following description relates to an electric motor and a motor-driven compressor.

#### 2. Description of Related Art

**[0003]** When a motor is driven by sensorless control, the position of a rotor of the motor needs to be estimated by a controller. Sensorless control is a control method that drives a motor without using a hardware position sensor and estimates the position of a rotor using software.

**[0004]** A typical method for estimating the position of a rotor includes, for example, an induced voltage method and a harmonic superposition method. The induced voltage method estimates the position of a rotor based on an induced voltage generated by rotation of a motor. Japanese Laid-Open Patent Publication No. 2011-172324 describes a harmonic superposition method that estimates the position of a rotor based on a current (response current) flowing through a motor when a harmonic component is superposed on a voltage instruction value or a current instruction value to the motor.

### SUMMARY

**[0005]** A motor loss may increase as the frequency of a carrier (carrier wave) for pulse width modulation (PWM) control of the motor and the frequency of a harmonic superposed for position estimation of a rotor become higher. In contrast, a decrease in the frequency of the harmonic may adversely affect the noise and vibration (NV) characteristics. Nonetheless, the above publication is mute on specific settings of the frequency of the harmonic.

**[0006]** This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

**[0007]** In one general aspect, an electric motor includes a motor unit, an inverter, a temperature-associated value obtainer, and processing circuitry. The motor unit includes a rotor and a stator around which three-phase coils are wound. The inverter includes a driver and a switching element operated by the driver. The inverter is configured to drive the motor unit through operation of the switching element. The temperature-associated value obtainer is configured to obtain a temperature-associated value that is associated with a temperature of the motor unit. The processing circuitry is configured to calculate instruction values for control of the switching element and control the switching element. The

processing circuitry is configured to superpose a harmonic on at least one of the instruction values. The processing circuitry is configured to estimate a position of the rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values. A frequency of the harmonic includes a first frequency and a second frequency that is lower than the first frequency. The processing circuitry is further configured to determine whether the temperature-associated value becomes greater than a threshold value. The processing circuitry is further configured to set the frequency of the harmonic for superposition on the at least one of the instruction values to the second frequency when the temperature-associated value becomes greater than the threshold value.

**[0008]** In another general aspect, a motor-driven compressor includes a compression unit configured to compress a fluid, and the above-described electric motor configured to drive the compression unit.

**[0009]** Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

### BRIEF DESCRIPTION OF THE DRAWINGS

**[0010]** FIG. 1 is a diagram showing an example of the configuration of a vehicle air conditioner.

**[0011]** FIG. 2 is a graph illustrating the relationship of a frequency of a harmonic switched by a controller of the vehicle air conditioner shown in FIG. 1 and an effective value of a phase current.

**[0012]** FIG. 3 is a flowchart illustrating an example of the series of processes executed by the controller of the vehicle air conditioner shown in FIG. 1 to control a motor unit.

**[0013]** FIG. 4 is a flowchart illustrating an example of the series of processes executed by the controller of the vehicle air conditioner shown in FIG. 1 to set the frequency of the harmonic.

**[0014]** FIG. 5 is a graph illustrating the relationship of a frequency of a harmonic switched by a controller of a modified example and an effective value of a phase current.

**[0015]** FIG. 6 is a flowchart illustrating an example of the series of processes executed by the controller of the modified example to set the frequency of the harmonic.

**[0016]** Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

### DETAILED DESCRIPTION

**[0017]** This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

**[0018]** Exemplary embodiments may have different forms, and are not limited to the examples described.

However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0019] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

#### Embodiment

[0020] An embodiment of an electric motor and a motor-driven compressor will now be described with reference to the drawings.

#### Overall Configuration

[0021] As shown in FIG. 1, a vehicle air conditioner 100 of the present embodiment includes a motor-driven compressor 101 and a refrigerant circuit 103. The motor-driven compressor 101 includes a compression unit 102 and an electric motor M1. The motor-driven compressor 101 compresses refrigerant. The refrigerant circuit 103 includes, for example, a heat exchanger, an expansion valve, or the like. The motor-driven compressor 101 compresses refrigerant, and the refrigerant circuit 103 exchanges heat with the refrigerant and expands the refrigerant. This allows the vehicle air conditioner 100 to cool or warm the passenger compartment. The motor-driven compressor 101 discharges oil together with the compressed refrigerant.

[0022] The compression unit 102 compresses refrigerant. The refrigerant compressed by the compression unit 102 is discharged to the refrigerant circuit 103. The compression unit 102 may be of any type, such as a scroll type, a piston type, a vane type, or the like.

#### Electric Motor

[0023] The electric motor M1 includes a motor unit 11. The motor unit 11 includes a rotor 12 and a stator 13 around which three-phase coils U, V, and W are wound. The motor unit 11 is a three-phase motor including three coils U, V, and W. The motor unit 11 drives the compression unit 102. The motor unit 11 may be cooled by the refrigerant circulated through the refrigerant circuit 103.

[0024] The electric motor M1 includes a motor driving device 10. The motor driving device 10 includes a battery BA, a smoothing capacitor C, an inverter 21, a phase current detector 22, an input voltage detector 23, and a controller 30.

[0025] The inverter 21 includes six switching elements Q1 to Q6, diodes D1 to D6, and a driver 51. The switching elements Q1 to Q6 are, for example, insulated gate bipolar transistors (IGBTs). When the switching elements Q1 to Q6 are metal-oxide-semiconductor field-effect transistors (MOSFETs), the switching elements Q1 to Q6 are integrated with the diodes D1 to D6. The switching elements Q1 and Q2 are connected in series to each other. The switching elements Q3 and Q4 are connected in series to each other. The switching elements Q5 and Q6 are connected in series to each other. The switching elements Q1 to Q6 are connected in parallel to the diodes D1 to D6, respectively. The switching elements Q1 to Q6 are each connected to the battery BA via the smoothing capacitor C.

[0026] A connection line connecting the switching elements Q1 and Q2 branches off and is connected to the coil U. A connection line connecting the switching elements Q3 and Q4 branches off and is connected to the coil V. A

connection line connecting the switching elements Q5 and Q6 branches off and is connected to the coil W.

[0027] The driver 51 operates the switching elements Q1 to Q6. The operation of the switching elements Q1 to Q6 drives the motor unit 11. The operation of the switching elements Q1 to Q6 refers to switching operations of the switching elements Q1 to Q6, that is, switching of the switching elements Q1 to Q6 between on and off states.

[0028] The battery BA is a power storage device that can be charged and discharged. The rated voltage of the battery BA is, for example, 800V.

[0029] The phase current detector 22 detects a phase current flowing through the motor unit 11. The phase current detector 22 detects phase currents of at least two phases. In the present embodiment, the phase current detector 22 detects a u-phase current I<sub>u</sub>, a v-phase current I<sub>v</sub>, and a w-phase current I<sub>w</sub>. In another embodiment, the phase current detector 22 may detect phase currents of two phases and calculate a phase current of the remaining phase from the two detected phase currents, taking consideration that the sum of phase currents of the three phases equals zero. The u-phase current I<sub>u</sub>, the v-phase current I<sub>v</sub>, and the w-phase current I<sub>w</sub> are actual currents flowing through the coils U, V, and W of the motor unit 11.

[0030] The input voltage detector 23 detects an input voltage V<sub>i</sub> applied to the inverter 21 by the battery BA.

#### Controller

[0031] The controller 30 includes a processor and memory 80. The processor is, for example, a central processing unit (CPU), a graphics processing unit (GPU), or a digital signal processor (DSP). The memory 80 includes a random-access memory (RAM) and a read-only memory (ROM). The memory 80 stores program codes or instructions configured to cause the processor to execute processes. The memory 80, which is a computer-readable medium, includes any medium accessible by a general-purpose computer or a special-purpose computer. The controller 30 may include a hardware circuit, such as an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or the like. The controller 30, which is processing circuitry, may include one or more processors that operate according to computer programs, one or more hardware circuits (e.g., ASIC or FPGA), or a combination of the above.

[0032] The controller 30 calculates instruction values. The controller 30 uses the instruction values to control the switching elements Q1 to Q6. The controller 30 drives the motor unit 11 by controlling the inverter 21. In the present embodiment, the controller 30 performs sensorless control that drives the motor unit 11 without using a hardware position sensor that detects a position H<sub>m</sub> of the rotor 12 of the motor unit 11. The controller 30 controls the inverter 21 while switching between position estimation using an induced voltage method and position estimation using a harmonic superposition method. The induced voltage method estimates the position H<sub>m</sub> of the rotor 12 based on an induced voltage generated in the three-phase coils U, V, and W. The harmonic superposition method estimates the position H<sub>m</sub> of the rotor 12 based on an actual current (response current) flowing through the motor unit 11 when a harmonic is superposed on an instruction value. Such control on the inverter 21 drives the motor unit 11. Position estimation by the induced voltage method and position estimation by the harmonic superposition method will now

be described. Details of a process executed by the controller 30 to switch the two methods will be described later.

#### Position Estimation by Induced Voltage Method

[0033] The functionality of the controller 30 when performing position estimation by the induced voltage method is described below. The controller 30 includes a current coordinate converter 31, a position estimator 32, subtractors 33, 35, and 36, a speed controller 34, a current controller 37, and a PWM controller 38.

[0034] The current coordinate converter 31 converts phase currents  $I_u$ ,  $I_v$ , and  $I_w$  to a d-axis current  $I_d$  and a q-axis current  $I_q$  based on the position  $H_m$  of the rotor 12 estimated by the position estimator 32. For example, the current coordinate converter 31 converts the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  of a three-phase (U, V, and W) fixed coordinate system into currents  $I_\alpha$  and  $I_\beta$  of a two-phase ( $\alpha$  and  $\beta$ ) fixed coordinate system. The current coordinate converter 31 uses the position  $H_m$  to convert the currents  $I_\alpha$  and  $I_\beta$  into the d-axis current  $I_d$  and the q-axis current  $I_q$  of a two-phase (d and q) rotating coordinate system. The d-axis and the q-axis are coordinate axes of a d-q coordinate system that rotates together with the rotor 12 of the motor unit 11. The current coordinate converter 31 may directly convert the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  to the d-axis current  $I_d$  and the q-axis current  $I_q$ , without converting them to the currents  $I_\alpha$  and  $I_\beta$ .

[0035] The position estimator 32 estimates the position  $H_m$  of the rotor 12 of the motor unit 11 based on the d-axis current  $I_d$  and the q-axis current  $I_q$  output from the current coordinate converter 31, and a d-axis voltage instruction value  $V_d$  and a q-axis voltage instruction value  $V_q$  output from the current controller 37. For example, the position estimator 32 calculates an induced voltage generated in the coils U, V, and W based on the d-axis current  $I_d$  and the q-axis current  $I_q$ , the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$  obtained from the current controller 37, and a constant determined by the motor unit 11. Then, the position estimator 32 estimates the position  $H_m$  based on the induced voltage. Further, the position estimator 32 estimates a rotation speed  $F_m$  of the rotor 12 based on the estimated position  $H_m$ . To facilitate understanding, arrows indicating the d-axis current  $I_d$ , the d-axis voltage instruction value  $V_d$ , and the q-axis voltage instruction value  $V_q$  input to the position estimator 32 are not shown in FIG. 1.

[0036] The position estimator 32 uses the induced voltage method to estimate the position of the rotor 12 while the estimated rotation speed  $F_m$  is greater than or equal to a predetermined rotation speed  $F_{mth}$ . The position estimator 32 uses the harmonic superposition method to estimate the position of the rotor 12 while the estimated rotation speed  $F_m$  is less than the predetermined rotation speed  $F_{mth}$ . For example, the predetermined rotation speed  $F_{mth}$  is a value indicating a rotation speed of the rotor 12 that allows for position estimation of rotor 12 by the induced voltage method. The predetermined rotation speed is, for example, approximately 700 rpm.

[0037] The subtractor 33 calculates a difference  $\Delta F_m$  between the rotation speed  $F_m$  estimated by the position estimator 32 and a rotation speed instruction value  $F_{mRef}$ . The rotation speed instruction value  $F_{mRef}$  is an instruction value of the rotation speed of the rotor 12 input from outside.

For example, the controller 30 receives the rotation speed instruction value  $F_{mRef}$  from a higher-level vehicle control device.

[0038] The speed controller 34 calculates a d-axis current instruction value  $I_{dRef}$  and a q-axis current instruction value  $I_{qRef}$  based on the difference  $\Delta F_m$ . For example, the speed controller 34 performs feedback control to calculate the d-axis current instruction value  $I_{dRef}$  and the q-axis current instruction value  $I_{qRef}$  so that the difference  $\Delta F_m$  converges to zero. The feedback control is, for example, proportional-integral control.

[0039] The subtractor 35 calculates a difference  $\Delta I_d$  between the d-axis current instruction value  $I_{dRef}$  and the d-axis current  $I_d$ . The subtractor 36 calculates a difference  $\Delta I_q$  between the q-axis current instruction value  $I_{qRef}$  and the q-axis current  $I_q$ .

[0040] The current controller 37 calculates the d-axis voltage instruction value  $V_d$  based on the difference  $\Delta I_d$ . The current controller 37 calculates the q-axis voltage instruction value  $V_q$  based on the difference  $\Delta I_q$ . For example, the current controller 37 performs feedback control to calculate the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$  so that the difference  $\Delta I_d$  and the difference  $\Delta I_q$  converge to zero. The feedback control is, for example, proportional-integral control.

[0041] The PWM controller 38 converts the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$  to a u-phase voltage instruction value  $V_u$ , a v-phase voltage instruction value  $V_v$ , and a w-phase voltage instruction value  $V_w$  based on the position  $H_m$  of the rotor 12 estimated by the position estimator 32 and the input voltage  $V_i$ . For example, the PWM controller 38 converts the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$  of the d-q coordinate system to voltage instruction values  $V_\alpha$  and  $V_\beta$  of a  $\alpha$ - $\beta$  coordinate system. The PWM controller 38 converts the two-phase voltage instruction values  $V_\alpha$  and  $V_\beta$  to three-phase voltage instruction values  $V_u$ ,  $V_v$ , and  $V_w$ . The PWM controller 38 may directly convert the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$  to the voltage instruction values  $V_u$ ,  $V_v$ , and  $V_w$ , without converting them to the voltage instruction values  $V_\alpha$  and  $V_\beta$ .

[0042] The inverter 21 is controlled based on the voltage instruction values  $V_u$ ,  $V_v$ , and  $V_w$ . Specifically, the PWM controller 38 calculates a duty instruction value based on the voltage instruction values  $V_u$ ,  $V_v$ , and  $V_w$ . Subsequently, the PWM controller 38 compares the duty instruction value with a carrier (carrier wave), such as a triangular wave, a sawtooth wave, or the like, to generate a PWM signal having a duty ratio that indicates the duty instruction value. The PWM signal is a gate-on-off signal that determines the ON-OFF states of the switching elements Q1 to Q6. The controller 30 controls the switching elements Q1 to Q6 by outputting the PWM signal to corresponding switching elements Q1 to Q6 via the driver 51.

#### Position Estimation by Harmonic Superposition Method

[0043] The functionality of the controller 30 when performing position estimation by the harmonic superposition method will now be described. As shown in FIG. 1, in addition to the configuration described above, the controller 30 further includes a harmonic superposer 41, an adder 42, a band-stop filter 43, and a harmonic superposition fre-



quency switcher **44**. The current coordinate converter **31**, the subtractors **33**, **35**, **36**, the speed controller **34**, the current controller **37**, and the PWM controller **38** have the same functionalities as in the induced voltage method. Thus, these components will not be described.

**[0044]** The harmonic superposer **41** generates a harmonic  $V_h$ . The harmonic  $V_h$  is defined by  $V_a \cdot \cos 2\pi f t$ . The “ $V_a$ ” represents the amplitude of the harmonic  $V_h$ , and the “ $f$ ” represents the frequency of the harmonic  $V_h$ . For example, the frequency  $f$  is not dependent on the rotation of the motor unit **11**.

**[0045]** The adder **42** adds the harmonic  $V_h$  generated by the harmonic superposer **41** to the d-axis voltage instruction value  $V_d$ . This superposes the harmonic  $V_h$  on the d-axis voltage instruction value  $V_d$ . In the present embodiment, the controller **30** superposes the harmonic  $V_h$  on only the d-axis voltage instruction value  $V_d$ . When the position estimator **32** estimates the position of the rotor **12** by the induced voltage method, the harmonic superposer **41** does not superpose the harmonic  $V_h$  on the d-axis voltage instruction value  $V_d$ . In another embodiment, the controller **30** may estimate the position of the rotor **12** by superposing the harmonic  $V_h$  on both the d-axis voltage instruction value  $V_d$  and the q-axis voltage instruction value  $V_q$ , both the d-axis current instruction value  $I_{dRef}$  and the q-axis current instruction value  $I_{qRef}$ , or a predetermined instruction value specified by three-phase voltage instruction values  $V_u$ ,  $V_v$ , and  $V_w$ .

**[0046]** The band-stop filter **43** removes a frequency component of a specific band from the d-axis current  $I_d$  and the q-axis current  $I_q$ . When the harmonic superposer **41** superposes the harmonic  $V_h$  on the d-axis voltage instruction value  $V_d$ , the d-axis current  $I_d$  and the q-axis current  $I_q$  contain frequency components based on the harmonic  $V_h$ . The band-stop filter **43** removes such frequency components.

**[0047]** As described above, the position estimator **32** estimates the position of the rotor **12** using the harmonic superposition method while the estimated rotation speed  $F_m$  is less than the predetermined rotation speed  $F_{mth}$ . Specifically, the position estimator **32** estimates the position  $H_m$  of the rotor **12** and the rotation speed  $F_m$  of the rotor **12** from the q-axis current  $I_q$  calculated by the current coordinate converter **31**. When the harmonic  $V_h$  is superposed on the d-axis voltage instruction value  $V_d$ , the q-axis current  $I_q$  contains a current harmonic based on the harmonic  $V_h$ . The position estimator **32** estimates the position  $H_m$  of the rotor **12** and the rotation speed  $F_m$  of the rotor **12** based on the current harmonic contained in the q-axis current  $I_q$  and a mathematical expression model of the motor unit **11**. For example, the position estimator **32** calculates an axis error  $\Delta\theta_c$  from the current harmonic. The axis error  $\Delta\theta_c$  is a difference between the actual position  $H_m$  of the rotor **12** and the position  $H_m$  of the rotor **12** recognized by the controller **30**. The position estimator **32** estimates the position  $H_m$  of the rotor **12** and the rotation speed  $F_m$  of the rotor **12** so that the axis error  $\Delta\theta_c$  becomes equal to zero.

**[0048]** The harmonic superposition frequency switcher **44** switches the frequency of the harmonic  $V_h$  based on the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  detected by the phase current detector **22** and a first threshold value  $Th1$ .

#### Relationship between Frequency of Harmonic $V_h$ and First Threshold Value $Th1$

**[0049]** FIG. 2 is a graph illustrating the relationship between the frequency of the harmonic  $V_h$  and an effective value of the phase current of the motor unit **11**. The frequency of the harmonic  $V_h$  includes a first frequency  $f1$  and a second frequency  $f2$ . The second frequency  $f2$  is lower than the first frequency  $f1$ . In the description hereafter, the first frequency  $f1$  may be referred to as the initial superposition frequency.

**[0050]** As shown in FIG. 2, while each of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  of the motor unit **11** is in a range of zero to the first threshold value  $Th1$ , the harmonic superposition frequency switcher **44** causes the harmonic superposer **41** to generate the harmonic  $V_h$  having the first frequency  $f1$  so that the harmonic  $V_h$  having the first frequency  $f1$  is superposed on the d-axis voltage instruction value  $V_d$ . When any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  detected by the phase current detector **22** becomes greater than the first threshold value  $Th1$ , the harmonic superposition frequency switcher **44** switches the frequency of the harmonic  $V_h$ , which is to be generated by the harmonic superposer **41**, from the first frequency  $f1$  to the second frequency  $f2$ . As a result, the harmonic  $V_h$  having the second frequency  $f2$  is superposed on the d-axis voltage instruction value  $V_d$ . The first threshold value  $Th1$  is based on the heat resistance temperature of the motor unit **11**. The first threshold value  $Th1$  indicates a phase current value when the temperature of the motor unit **11** is higher than the average temperature of the motor unit **11** in a normal state by a predetermined value.

**[0051]** As the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  of the motor unit **11** increase, the temperature of the motor unit **11** increases. That is, the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  of the motor unit **11** correlate with the temperature of the motor unit **11**. Therefore, the phase current detector **22** is an example of a temperature-associated value obtainer, and a detection result of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  is an example of a temperature-associated value that is associated with the temperature of the motor unit **11**. In other words, a process that determines whether any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  detected by the phase current detector **22** becomes greater than the first threshold value  $Th1$  is an example of a process that determines whether the temperature of the motor unit **11** becomes greater than the first threshold value  $Th1$ .

#### Process of Controller **30**

**[0052]** The series of processes executed by the controller **30** to control the motor unit **11** will now be described with reference to FIG. 3. The process of the flowchart shown in FIG. 3 is executed in predetermined calculation cycles from when the motor unit **11** is started to when the motor unit **11** is stopped. For example, the motor unit **11** is started when the vehicle air conditioner **100** is started, and is stopped when the vehicle air conditioner **100** is stopped.

**[0053]** In the series of processes related to the control of the motor unit **11**, the controller **30** first obtains the rotation speed  $F_m$  of the rotor **12** (step S1). In the initial calculation cycle after the motor unit **11** is started, the controller **30** may estimate the rotation speed  $F_m$  using the harmonic superposition method or the induced voltage method.

**[0054]** Next, the controller **30** determines whether the rotation speed  $F_m$  is greater than or equal to the predeter-

mined rotation speed  $F_{mth}$  (step S2). When the rotation speed  $F_m$  is greater than or equal to the predetermined rotation speed  $F_{mth}$  (step S2: YES), the controller 30 uses the induced voltage method to estimate the rotation position of the rotor 12 (step S3). When the rotation speed  $F_m$  is less than the predetermined rotation speed  $F_{mth}$  (step S2: NO), the controller 30 uses the harmonic superposition method to estimate the rotation position of the rotor 12 (step S4). Subsequently, the controller 30 generates a PWM signal based on the rotation speed  $F_m$  estimated in step S3 or step S4, and then controls the inverter 21 to drive the motor unit 11 (step S5).

[0055] Also, the controller 30 executes the series of processes illustrated in FIG. 4, simultaneously with the flowchart in FIG. 3, to set the frequency of the harmonic  $V_h$ . The process of the flowchart shown in FIG. 4 may be executed in the same calculation cycles as the flowchart in FIG. 3, or may be executed in different calculation cycles. The process of the flowchart shown in FIG. 4 does not have to be executed simultaneously with the flowchart in FIG. 3. For example, the process of the flowchart shown in FIG. 4 may be a subroutine executed within the flowchart in FIG. 3 (for example, executed before step S4).

[0056] When the motor unit 11 is started, the frequency of the harmonic  $V_h$  is set to the initial superposition frequency (first frequency  $f_1$ ). Therefore, if the motor unit 11 was stopped after the frequency of the harmonic  $V_h$  had been switched to the second frequency  $f_2$ , the frequency of the harmonic  $V_h$  is switched from the second frequency  $f_2$  to the initial superposition frequency when the motor unit 11 is started again.

[0057] In the series of processes related to the setting of the frequency of the harmonic  $V_h$ , the phase current detector 22 first obtains the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  (step S102). Next, the harmonic superposition frequency switcher 44 determines whether the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  are increased based on the detection result of the phase current detector 22 (step S104). When none of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  is increased (step S104: NO), there is no need to switch the frequency of the harmonic  $V_h$ . Thus, the harmonic superposition frequency switcher 44 ends the series of processes.

[0058] When the harmonic superposition frequency switcher 44 determines that any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  is increased (step S104: YES), the harmonic superposition frequency switcher 44 determines whether any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  has become greater than the first threshold value  $Th_1$  (step S106). When the harmonic superposition frequency switcher 44 determines that none of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  has become greater than the first threshold value  $Th_1$  (step S106: NO), the harmonic superposition frequency switcher 44 ends the series of processes. When the harmonic superposition frequency switcher 44 determines that any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  has become greater than the first threshold value  $Th_1$  (step S106: YES), the harmonic superposition frequency switcher 44 switches the frequency of the harmonic  $V_h$  from the first frequency  $f_1$  to the second frequency  $f_2$  (step S108). In this manner, the harmonic superposer 41 superposes the harmonic  $V_h$  having the second frequency  $f_2$  on the d-axis voltage instruction value  $V_d$ . Then, the controller 30 ends the series of processes.

#### Advantages of the Embodiment

[0059] The above embodiment has the following advantages.

[0060] (1) The position estimator 32 estimates the position of the rotor 12 by the harmonic superposition method. The frequency of the harmonic  $V_h$  is set to the first frequency  $f_1$  when the motor unit 11 is started. When any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  exceeds the first threshold value  $Th_1$ , the harmonic superposition frequency switcher 44 determines that the temperature of the motor unit 11 has become greater than the first threshold value. Then, the harmonic superposition frequency switcher 44 switches the frequency of the harmonic  $V_h$ , which is to be superposed on the d-axis voltage instruction value  $V_d$ , from the first frequency  $f_1$  to the second frequency  $f_2$ .

[0061] With this configuration, the frequency of the harmonic  $V_h$  is set relatively high when position estimation of the rotor 12 by the harmonic superposition method is started. This minimizes the adverse effects on the NV characteristics. Furthermore, the frequency of the harmonic  $V_h$  is lowered when the temperature of the motor unit 11 increases. This avoids overheating of the motor unit 11.

[0062] (2) The detection result of the phase current detector 22, which detects the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  flowing through the motor unit 11, is used as the temperature-associated value that is associated with the temperature of the motor unit 11.

[0063] This configuration obtains the temperature-associated value of the motor unit 11 without an additional configuration for detecting the temperature of the motor unit 11.

[0064] (3) The position estimator 32 estimates the position of the rotor 12 using the harmonic superposition method or the induced voltage method. Specifically, when the rotation speed  $F_m$  of the rotor 12 based on the estimation result is greater than or equal to the predetermined rotation speed  $F_{mth}$ , the position estimator 32 uses the induced voltage method to estimate the position of the rotor 12. When the rotation speed  $F_m$  of the rotor 12 based on the estimation result is less than the predetermined rotation speed  $F_{mth}$ , the position estimator 32 uses the harmonic superposition method to estimate the position of the rotor 12. When the position estimator 32 estimates the position of the rotor 12 by the induced voltage method, the harmonic superposer 41 does not superpose the harmonic  $V_h$  on the d-axis voltage instruction value  $V_d$ .

[0065] As the rotation speed  $F_m$  of the rotor 12 decreases, the induced voltage decreases. Therefore, if the rotation speed  $F_m$  of the rotor 12 is relatively low when the position estimator 32 estimates the position of the rotor 12 using the induced voltage method, the estimation accuracy may become lower. In this respect, the above configuration allows the position estimator 32 to accurately estimate the position of the rotor 12 using an appropriate one of the harmonic superposition method and the induced voltage method in accordance with the rotation speed  $F_m$  of the rotor 12. Furthermore, if the rotational speed  $F_m$  of the rotor 12 is relatively low when the refrigerant for cooling the motor unit 11 is circulated by the rotation of the motor unit 11, the flow rate of refrigerant may be reduced. This may cause overheating of the motor unit 11. Therefore, it is highly

advantageous to lower the frequency of the harmonic Vh when the temperature of the motor unit 11 is relatively high, as in the above embodiment.

[0066] (4) The motor-driven compressor 101 includes the compression unit 102 that compresses the refrigerant, which is a fluid, and the electric motor M1 that drives the compression unit 102. The frequency of the harmonic Vh is switched when estimating the position of the rotor 12 by the harmonic superposition method, as described above. This avoids overheating of the motor unit 11 and minimizes the adverse effects on the NV characteristics. With this configuration, the vehicle air conditioner 100 including the motor-driven compressor 101 avoids overheating of the motor unit 11 and reduces the noise and vibration experienced by occupants in the passenger compartment.

[0067] The above embodiment may be modified as follows. The above embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0068] The harmonic superposition frequency switcher 44 switches the frequency of the harmonic Vh from the first frequency f1 to the second frequency f2 when the phase currents Iu, Iv, and Iw of the motor unit 11 are increased. However, there is no limit to such a configuration. The harmonic superposition frequency switcher 44 may switch the frequency of the harmonic Vh from the second frequency f2 to the first frequency f1 when the phase currents Iu, Iv, and Iw of the motor unit 11 are decreased. The details will now be described with reference to FIGS. 5 to 6.

[0069] When any of the phase currents Iu, Iv, and Iw of the motor unit 11 becomes less than or equal to the first threshold value Th1, the harmonic superposition frequency switcher 44 maintains the frequency of the harmonic Vh at the second frequency f2 while the phase currents Iu, Iv, and Iw are greater than or equal to a second threshold value Th2. The second threshold value Th2 is less than the first threshold value Th1. When the frequency of the harmonic Vh is the second frequency f2, the harmonic superposition frequency switcher 44 switches the frequency of the harmonic Vh to the first frequency f1 after any of the phase currents Iu, Iv, and Iw of the motor unit 11 becomes less than the second threshold value Th2. As described above, if any of the phase currents Iu, Iv, and Iw of the motor unit 11 becomes greater than the first threshold value Th1 when the frequency of the harmonic Vh is the first frequency f1, the harmonic superposition frequency switcher 44 switches the frequency of the harmonic Vh to the second frequency f2. That is, hysteresis is provided in the switching of the frequency of the harmonic Vh. A process that determines whether any of the phase currents Iu, Iv, and Iw detected by the phase current detector 22 becomes less than the second threshold value Th2 is an example of a process that determines whether the temperature-associated value of the motor unit 11 becomes less than the second threshold value Th2.

[0070] The series of processes related to the setting of the frequency of the harmonic Vh in accordance with the modified example will now be described with reference to FIG. 6. The flowchart shown in FIG. 6 includes steps S200 to S202, in addition to the process of the flowchart in FIG. 4.

[0071] When the phase currents Iu, Iv, and Iw are not increased (step S104: NO), the harmonic superposition frequency switcher 44 determines that the phase currents Iu,

Iv, and Iw are decreased and determines whether any of the phase currents Iu, Iv, and Iw is less than the second threshold value Th2 (step S200). When the harmonic superposition frequency switcher 44 determines that none of the phase currents Iu, Iv, and Iw is less than the second threshold value Th2 (step S200: NO), the harmonic superposition frequency switcher 44 ends the series of processes. When the harmonic superposition frequency switcher 44 determines that any of the phase currents Iu, Iv, and Iw is less than the second threshold value Th2 (step S200: YES), the harmonic superposition frequency switcher 44 switches the frequency of the harmonic Vh from the second frequency f2 to the first frequency f1 (step S202). In this manner, the harmonic superposer 41 superposes the harmonic Vh having the first frequency f1 on the d-axis voltage instruction value Vd. Then, the controller 30 ends the series of processes.

[0072] This configuration appropriately switches the frequency of the harmonic Vh in accordance with the temperature of the motor unit 11. Further, hysteresis is provided in the switching of the frequency of the harmonic Vh, thereby avoiding frequent switching of the frequency of the harmonic Vh.

[0073] In the present modified example, the frequency of the harmonic Vh may be set to the second frequency f2 when the motor unit 11 is started.

[0074] The detection result of the phase currents Iu, Iv, and Iw detected by the phase current detector 22 is used as the temperature-associated value of the motor unit 11. However, there is no limit to such a configuration. Instead of (or in addition to) the detection result of the phase currents Iu, Iv, and Iw, other types of information may be used as the temperature-associated value of the motor unit 11. For example, in addition to the above configuration, the motor driving device 10 may include a temperature detector attached to the motor unit 11 and configured to detect the temperature of the motor unit 11. The temperature detector is implemented by, for example, a thermocouple and a functional unit that performs AD conversion on a measurement result of the thermocouple.

[0075] In this case, a detection result of the temperature detector is used as the temperature-associated value of the motor unit 11. The harmonic superposition frequency switcher 44 compares the temperature-associated value of the motor unit 11, which is based on the detection result of the temperature detector, with the first threshold value Th1 and the second threshold value Th2 to switch the frequency of the harmonic Vh. In this case, the first threshold value Th1 and the second threshold value Th2 are based on the heat resistance temperature of the motor unit 11. Specifically, the first threshold value Th1 indicates a temperature higher than the average temperature of the motor unit 11 in a normal state. The second threshold value Th2 is less than the first threshold value Th1.

[0076] The first threshold value Th1 does not have to be based on the average temperature of the motor unit 11 in a normal state. The first threshold value Th1 may be based on a statistical value, such as the median value, average value, mode value, or the like, of a temperature range achievable by the motor unit 11. Alternatively, the first threshold value Th1 may be based on a statistical value, such as the median value, average value, or the like, of a heat resistance temperature range of the motor unit 11.

[0077] The harmonic superposition frequency switcher 44 switches the frequency of the harmonic Vh based on the

effective value of any of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  detected by the phase current detector 22. However, there is no limit to such a configuration. The harmonic superposition frequency switcher 44 may switch the frequency of the harmonic  $V_h$  based on a statistical value, such as the median value, average value, maximum value, minimum value, or the like, of the phase currents  $I_u$ ,  $I_v$ , and  $I_w$  detected by the phase current detector 22.

[0078] The position estimator 32 estimates the position of the rotor 12 using the harmonic superposition method or the induced voltage method. However, there is no limit to such a configuration. The position estimator 32 may estimate the position of the rotor 12 using only the harmonic superposition method.

[0079] The harmonic superposer 41 superposes the harmonic  $V_h$  on the d-axis voltage instruction value  $V_d$ . However, there is no limit to such a configuration. The harmonic superposer 41 may superpose the harmonic  $V_h$  on the current instruction values  $I_{dRef}$  and  $I_{qRef}$ .

[0080] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

What is claimed is:

1. An electric motor comprising:

a motor unit including a rotor and a stator around which three-phase coils are wound;

an inverter including a driver and a switching element operated by the driver, the inverter being configured to drive the motor unit through operation of the switching element;

a temperature-associated value obtainer configured to obtain a temperature-associated value that is associated with a temperature of the motor unit; and

processing circuitry configured to calculate instruction values for control of the switching element and control the switching element, wherein

the processing circuitry is configured to:

superpose a harmonic on at least one of the instruction values; and

estimate a position of the rotor by a harmonic superposition method based on the harmonic superposed on the at least one of instruction values,

a frequency of the harmonic includes a first frequency and a second frequency that is lower than the first frequency, and

the processing circuitry is further configured to:

determine whether the temperature-associated value becomes greater than a threshold value; and

set the frequency of the harmonic for superposition on the at least one of the instruction values to the second frequency when the temperature-associated value becomes greater than the threshold value.

2. The electric motor according to claim 1, wherein the frequency of the harmonic is set to the first frequency when the motor unit is started.

3. The electric motor according to claim 1, wherein the threshold value is a first threshold value, and a second threshold value is less than the first threshold value, and the processing circuitry is further configured to:

determine whether the temperature-associated value becomes less than the second threshold value when the frequency of the harmonic is the second frequency; and

set the frequency of the harmonic to the first frequency when the temperature-associated value becomes less than the second threshold value.

4. The electric motor according to claim 1, wherein the temperature-associated value obtainer includes a phase current detector configured to detect a phase current flowing through the motor unit, and

the temperature-associated value includes a detection result of the phase current detector.

5. The electric motor according to claim 1, wherein the temperature-associated value obtainer includes a temperature detector attached to the motor unit and configured to detect the temperature of the motor unit, and the temperature-associated value includes a detection result of the temperature detector.

6. The electric motor according to claim 1, wherein the processing circuitry is further configured to:

estimate the position of the rotor by an induced voltage method based on an induced voltage generated in the three-phase coils when a rotation speed of the rotor is greater than or equal to a predetermined rotation speed; and

estimate the position of the rotor by the harmonic superposition method when the rotation speed is less than the predetermined rotation speed, and

when the position of the rotor is estimated by the induced voltage method, the processing circuitry is further configured to not perform superposition of the harmonic.

7. A motor-driven compressor, comprising:

a compression unit configured to compress a fluid; and the electric motor according to claim 1 and configured to drive the compression unit.

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